

AUSTRALASIAN SCHOOLS

SCIENCE

COMPETITION 2003

YEAR
5

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

Read the instructions on the **ANSWER SHEET**
and fill in your **NAME, SCHOOL YEAR, GENDER**
and the **LANGUAGE YOU FIRST SPOKE**.

Use the information provided to choose the **BEST** answer
from the four possible options.

On your **ANSWER SHEET** blacken the oval that matches
the answer you choose for each question.
Mark only **ONE** answer for each question.

Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

Use a 2B or B pencil.
Do **NOT** use a biro or pen.
Rub out any mistakes completely.
You may use a calculator and a ruler.

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EDUCATIONAL
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- 1 The photos show the actual size of four small rocks.



What is the length of the longest rock?

- (A) $2\frac{1}{2}$ cm (B) 3 cm (C) $3\frac{1}{2}$ cm (D) 4 cm

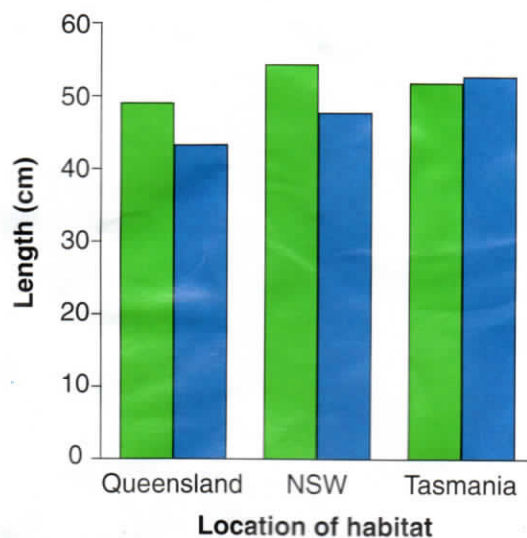
- 2 The table provides information about some animals.

Animal	Has 6 legs	Has 8 legs	Has wings	Lives in the sea
grasshopper	✓		✓	
honey bee	✓		✓	
lobster		✓		✓
spider		✓		

Which animal has 8 legs and lives in the sea?

- (A) grasshopper (B) honey bee (C) lobster (D) spider

- 3 The graph compares the lengths of some platypuses.



Which platypus is the longest?

- (A) female in Queensland
(B) female in Tasmania
(C) male in Tasmania
(D) male in NSW

- 4 Some students carried out tests on four powders.

They were told that a powder is:

- soluble in water if it dissolves to form a clear mixture
- not soluble in water if the mixture stays cloudy

The test results are shown in the table.

Powder	Test	
	Do the powder and water form a clear mixture?	What colour does red cabbage juice turn when the powder is placed in it?
cornflour	no	blue
baking soda	yes	blue
washing soda	yes	green
epsom salts	yes	blue

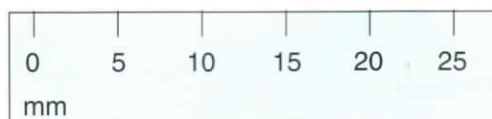
Which powder was **not** soluble?

- (A) cornflour
- (B) baking soda
- (C) washing soda
- (D) epsom salts

- 5 The photograph of the bumble bee and the ruler are shown at the same scale.



Photograph © 1999 Laura Smith



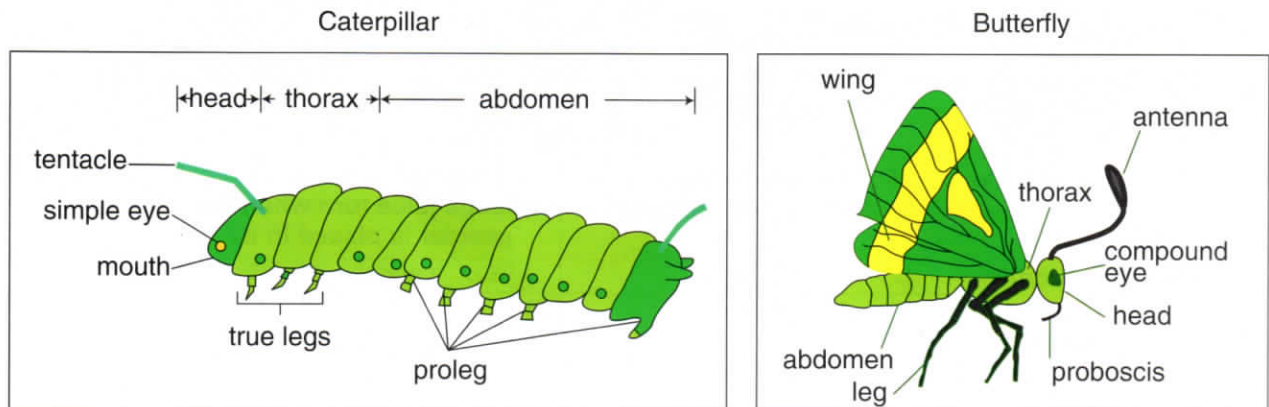
What is the length of the bumble bee from head to tail?

- (A) 14 mm (B) 18 mm (C) 40 mm (D) 50 mm

6

Insects have three main body parts: head, thorax and abdomen.

The drawings show the body parts of one side of a caterpillar and a butterfly.

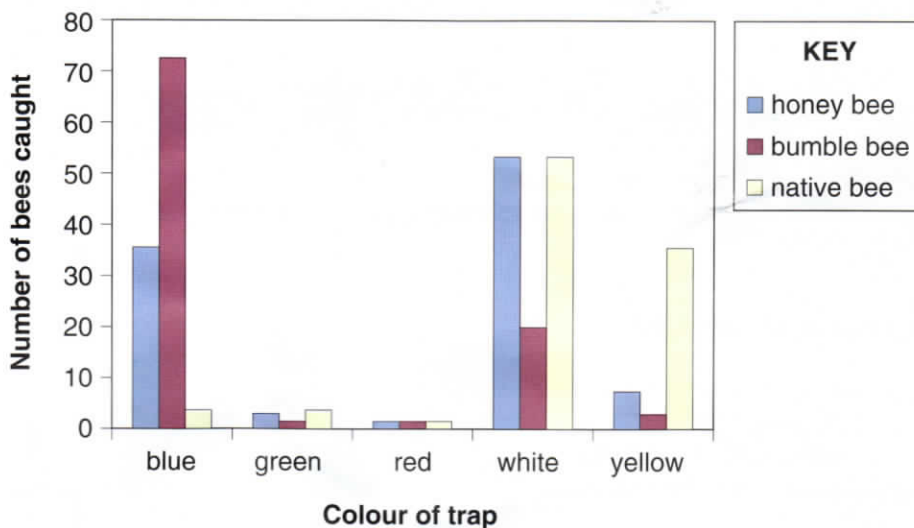


A caterpillar changed into a butterfly. What did **not** change about the insect?

- (A) type of eyes
- (B) type of mouth
- (C) number of prolegs
- (D) number of main body parts

7

The graph shows the number of each type of bee caught in different coloured traps during an experiment.



What was the colour of the trap in which most of the honey bees were caught?

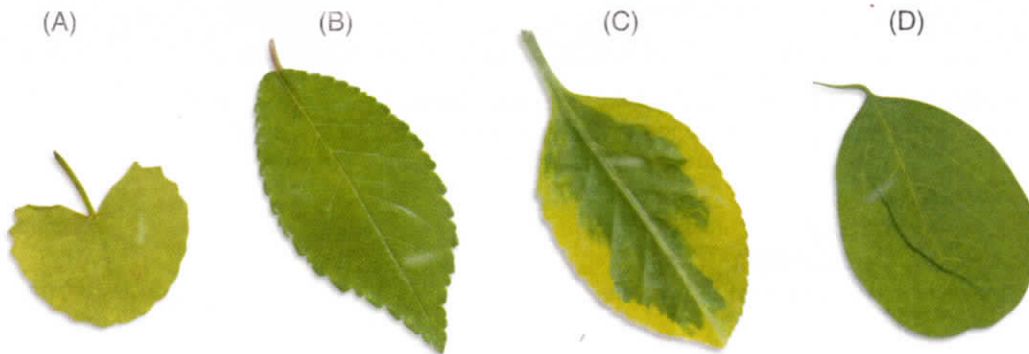
- (A) blue
- (B) green
- (C) white
- (D) yellow

8

Max took a photo of four leaves. He answered questions about each leaf.

Leaf	What is the leaf's shape?	Does the leaf have grooved edges?	What colour is the leaf?
P	blade	yes	dark green
Q	blade	yes	light green and dark green
R	oval	no	dark green
S	heart	yes	light green

Which photo shows leaf P?



9

Starfish live in rock pools on the coast. Naomi recorded the type of food some starfish ate.

Food type	Plant or animal	Number of starfish eating food type
algae	plant	21
marine snail	animal	6
prawns	animal	3

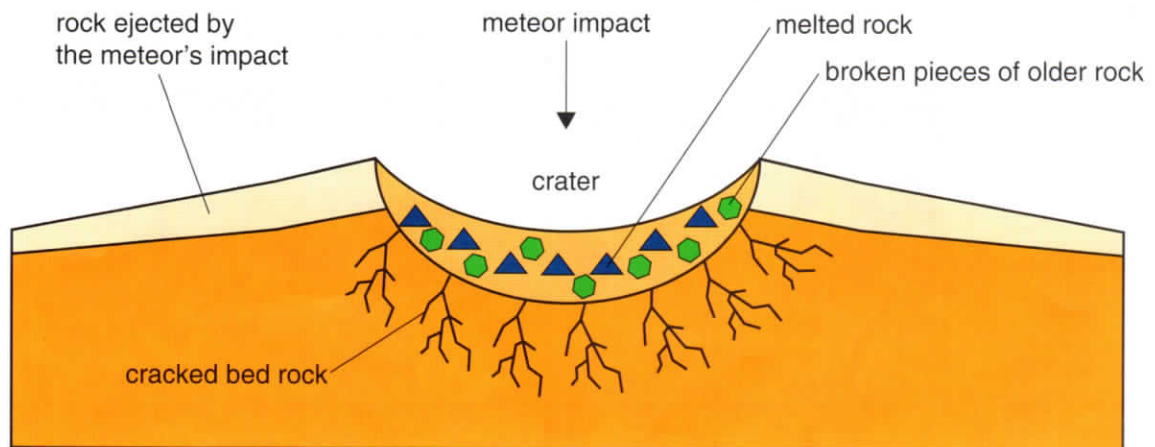
What could Naomi conclude from her observations?

- (A) The starfish ate both plants and animals.
- (B) The starfish ate plants only.
- (C) More starfish ate prawns than marine snails.
- (D) More starfish ate marine snails than algae.

10

When a meteor from outer space hits the moon, its impact makes a large crater.

The drawing gives information about a crater on the moon.

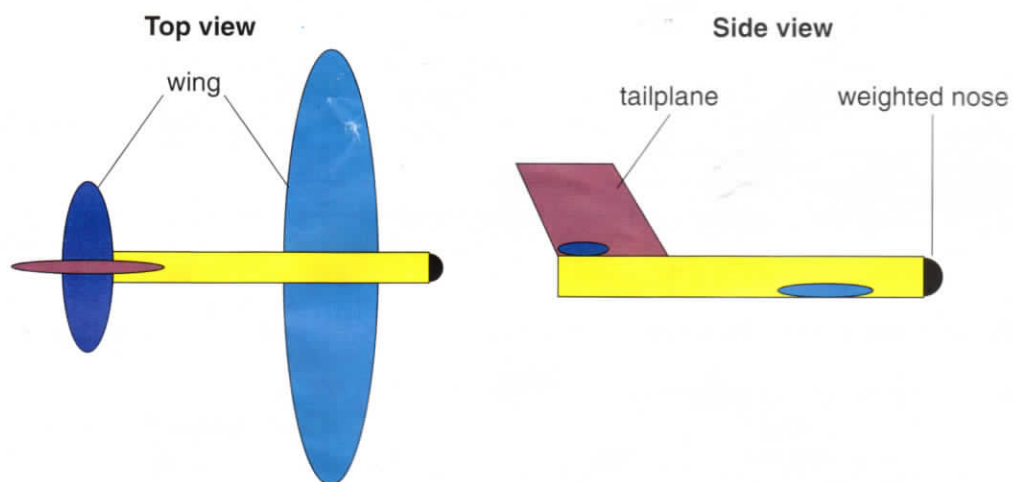


According to the diagram, what type of rock would be found on the crater's surface?

- (A) ejected rock
- (B) cracked bed rock
- (C) melted rock and ejected rock
- (D) melted rock and broken older rock

11

The diagrams, drawn to the same scale, show two views of a model plane.

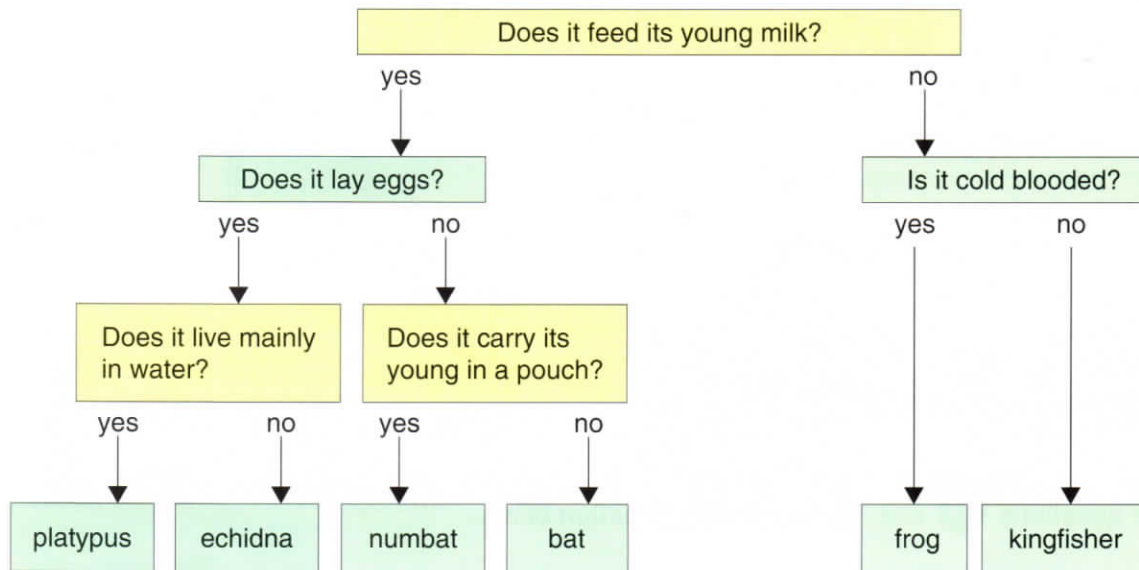


Which diagram shows the front view of the plane correctly?



12

The flowchart distinguishes between some Australian animals.



According to the flowchart, how are a platypus and a bat the same?

Both animals

- (A) carry their young in a pouch.
- (B) feed their young milk.
- (C) lay eggs.
- (D) live mainly in water.

13

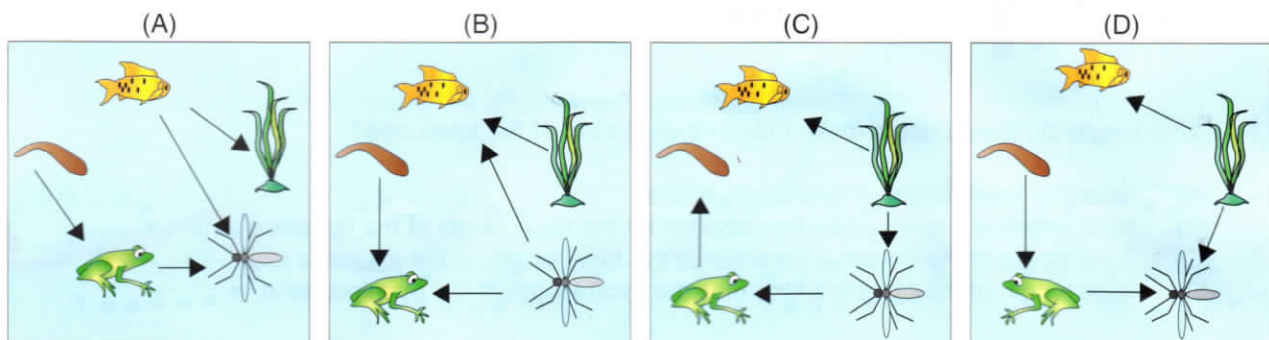
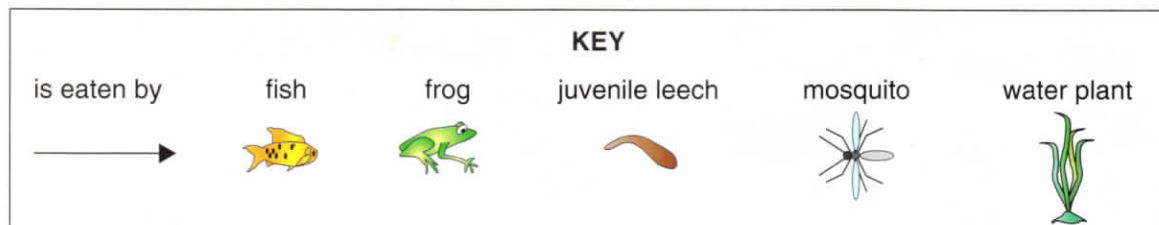
A food web shows how energy passes from one organism to another as it is eaten.

For example: "water plant $\longrightarrow$ fish"

means that energy passes from the water plant to the fish when the fish eats the plant.

The habitat of a leech is in and around fresh water ponds with muddy bottoms. Juvenile leeches feed on frogs until their jaws are strong enough to penetrate the skin of mammals.

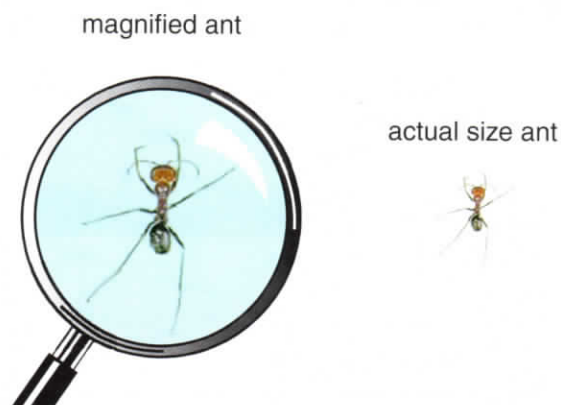
Which food web correctly shows the position of the juvenile leech?



- 14 An ant is shown at its actual size and as it appeared through a magnifying glass.

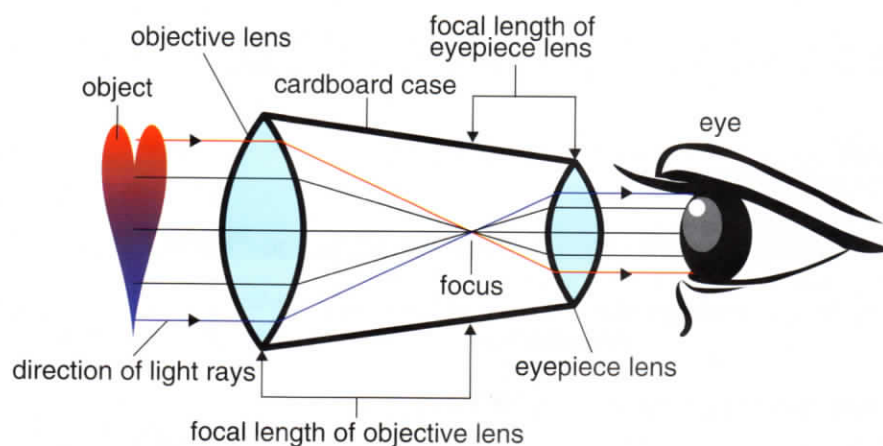
How many times larger did the ant appear through the magnifying glass?

- (A) half as large
- (B) one and a half times as large
- (C) twice as large
- (D) two and a half times as large

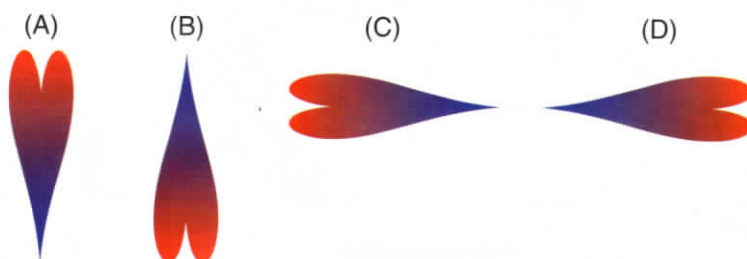


For questions 15 and 16 use the information below.

Some students made a telescope using two lenses. The cross-sectional diagram shows how the students constructed the telescope. It also shows how light from an object travels through the telescope to the eye.



- 15 How would the image of the object look through the telescope?



- 16 What length did the students make the cardboard case of the telescope?

- (A) twice the focal length of the objective lens
- (B) focal length of the objective lens divided by the focal length of the eyepiece lens
- (C) focal length of the objective lens minus the focal length of the eyepiece lens
- (D) focal length of the objective lens plus the focal length of the eyepiece lens

- 17 Tom thought that most vehicles entering the street beside his school were taking students to and from school. The school day was from 9 am to 3 pm.

He counted the number of vehicles that entered the street in one school day. His results are shown in the table.

Type of vehicle	Time							
	8 – 9 am	9 – 10 am	10 – 11 am	11 – 12 am	12 – 1 pm	1 – 2 pm	2 – 3 pm	3 – 4 pm
push bikes	54	0	0	0	0	0	3	51
cars	110	10	12	2	15	13	40	120
bus	8	1	1	1	1	1	1	8
trucks	0	2	2	1	1	0	0	0
motor bikes	0	0	1	0	0	1	0	0

Which vehicle numbers support Tom's idea?

- (A) cars only
- (B) cars and buses only
- (C) push bikes, cars, buses and trucks only
- (D) push bikes, cars, buses, trucks and motor bikes

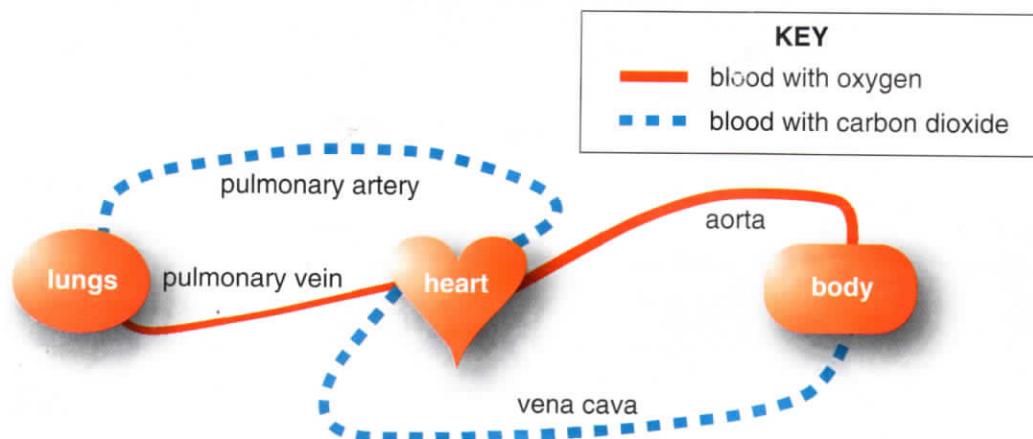
- 18 The circulatory system carries blood containing the gases, oxygen and carbon dioxide, through blood vessels to all parts of the body.

In the heart, blood is pumped to the blood vessels without changing it.

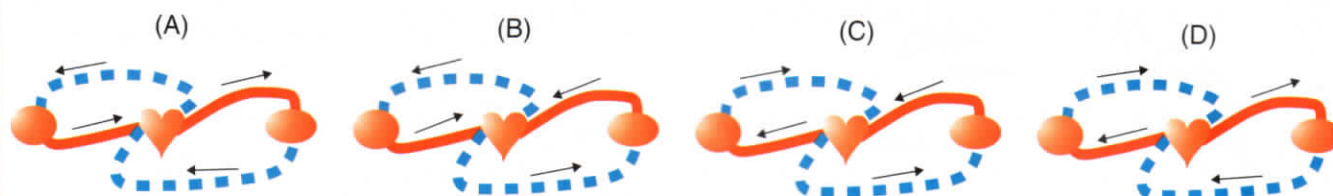
In the lungs, blood gives out carbon dioxide and takes up oxygen.

In the body, oxygen from the blood is used up and carbon dioxide from the body is taken up by the blood.

The diagram shows the parts of the circulatory system.



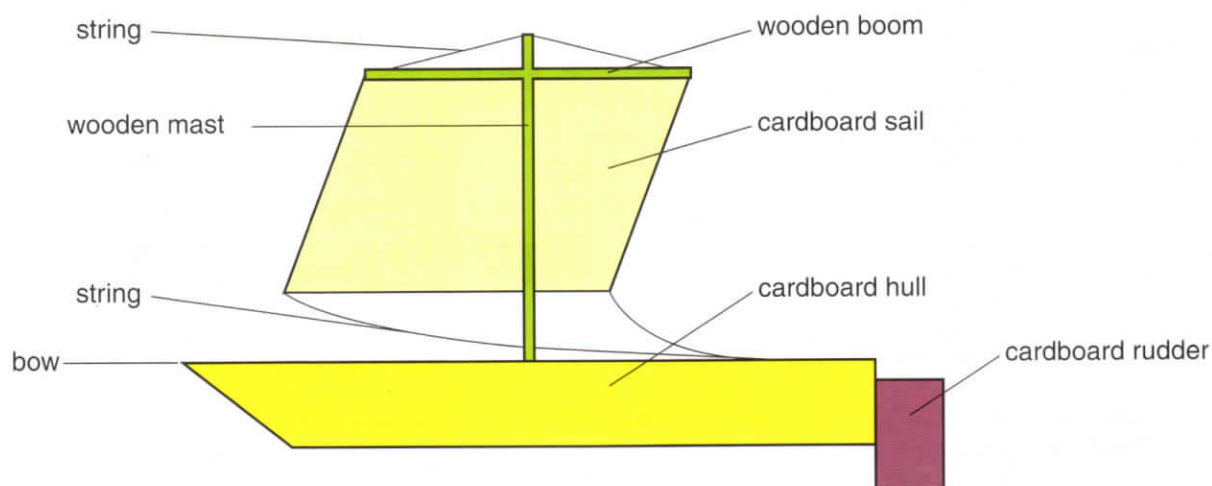
Which diagram correctly shows the direction in which blood flows?



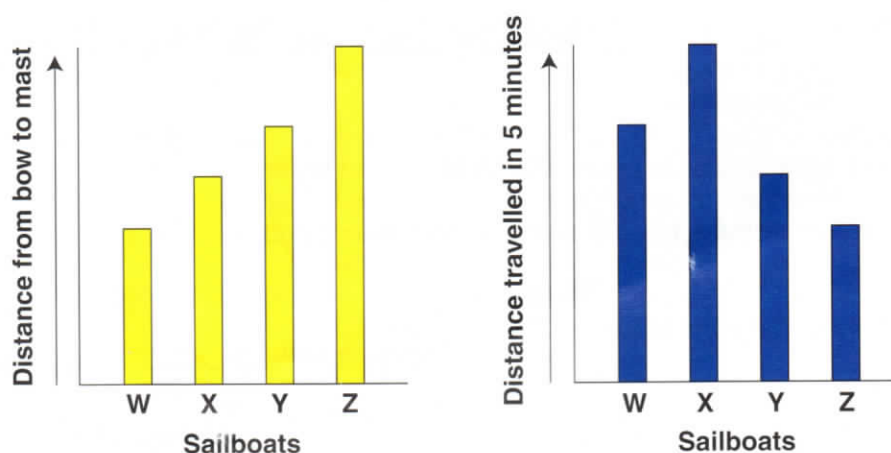
For questions 19 and 20 use the information below.

Some students wanted to know what effect the position of the mast had on the speed of a sailboat.

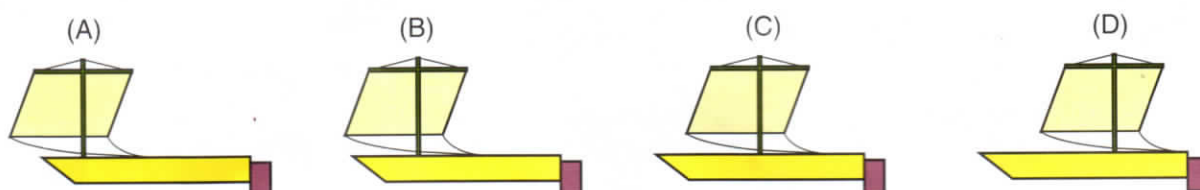
They made four model sailboats (W, X, Y and Z) using the design shown. The sailboats were the same except for the positions of their masts. Each mast was placed a different distance from the bow.



The graphs show the distances from the bow to the mast for each of the boats and the distances each boat travelled in 5 minutes.



- 19 Which diagram shows the mast position for the fastest sailboat?



- 20 The students thought that changing the size of the cardboard sail would affect the boat speeds.

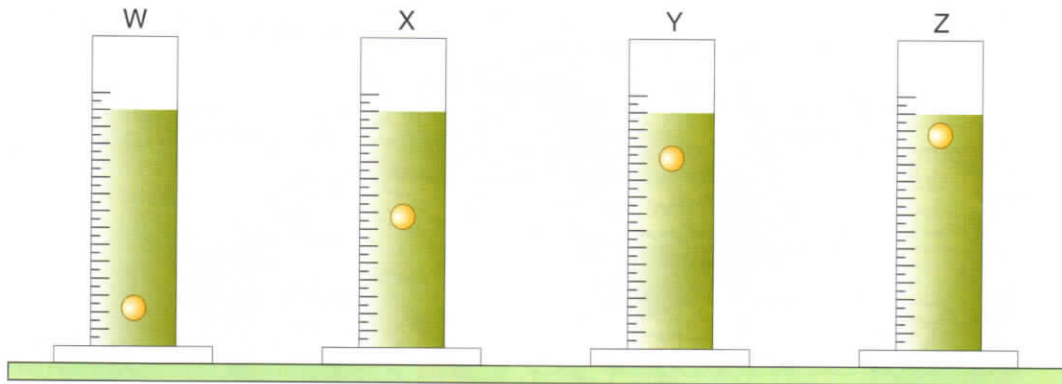
Which of the three slower boats would then need the least change to its sail size to reach the same speed as the unchanged fastest boat?

- (A) W (B) X (C) Y (D) Z

21

Viscosity is a measure of a liquid's thickness and stickiness. The more viscous the liquid, the longer it takes for an object to pass through it. Generally, the viscosity of a liquid decreases as temperature increases.

The diagram shows the distances clay balls of the same size pass through four motor oils in the same time. The four oils were at the same temperature.



When cold, high viscosity motor oil may not get to the part of the engine it is supposed to protect. When hot, low viscosity oil may not protect engine parts because it does not stick to them.

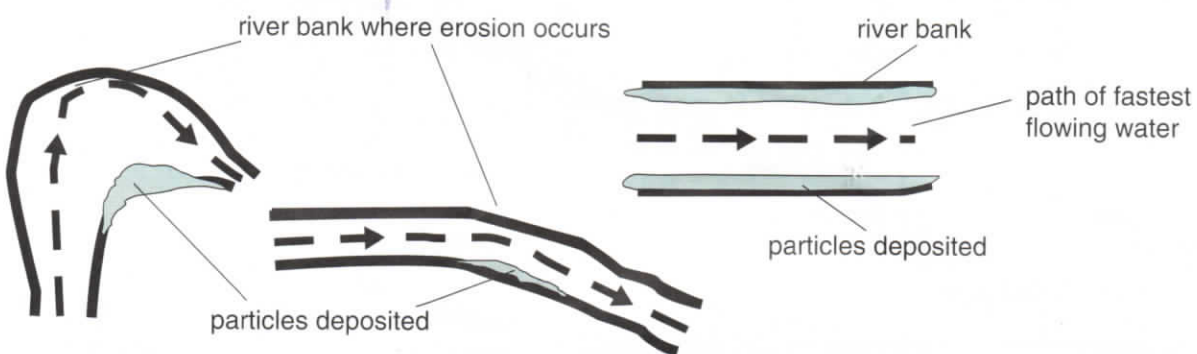
Which motor oil would be best for protecting engine parts at high temperatures?

- (A) W (B) X (C) Y (D) Z

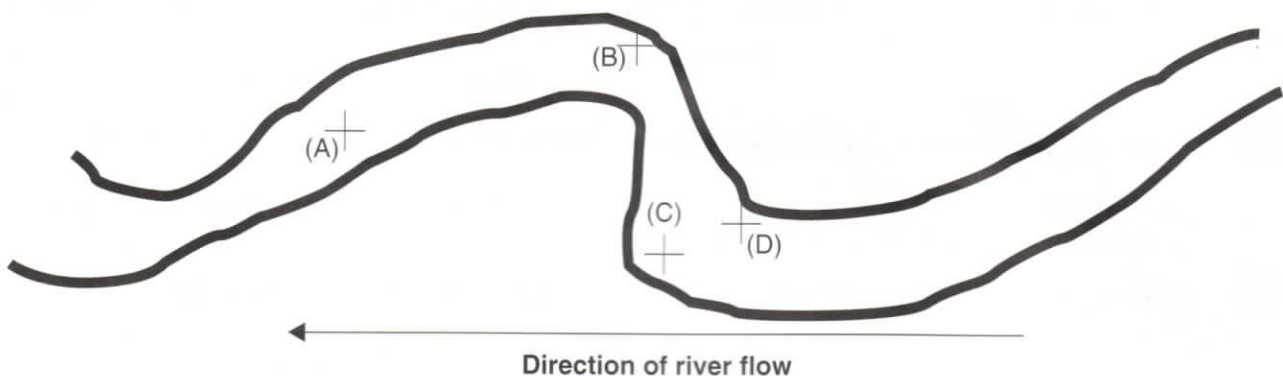
22

Fast moving water in a river erodes (carries away) the rock surfaces over which it flows. Slow moving water in a river deposits (drops) the rock particles it is carrying.

The diagrams show erosion and deposition occurring in some portions of a river.



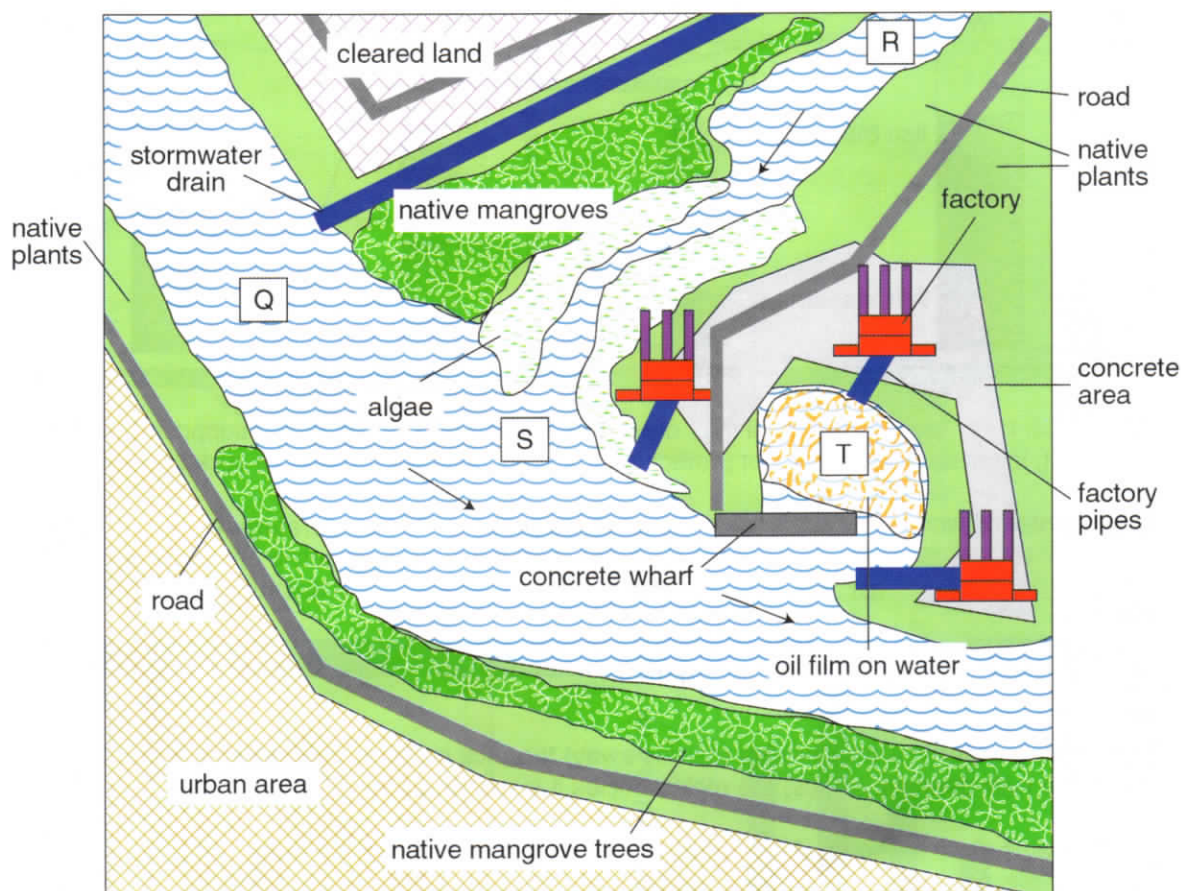
The diagram below shows another section of river.
At which position in the river would particles most likely be deposited?



For questions 23 and 24 use the information below.

Some students were asked to rate the health of a local river.

The map shows what they found in and around the river. The arrows on the map indicate the direction of water flow.



The students rated the health of the river as poor in five of the six categories.

River health category	Indicators of good health	Indicators of poor health
1. Land use	river bank is natural and undisturbed	river bank land has been cleared, factories present
2. Litter	river in natural state	plastic and large growths of algae in river
3. Pipes and drains	no pipes or drains enter river	pipes from factories and stormwater drains enter river
4. Structures	no structures, no changes to water flow	large structures, changes to water flow
5. Vegetation	mainly native plants growing on river bank	many non-native plants growing on river bank
6. Water clarity	water in river is colourless	water in river is brown or green, oil film on top of water

23 For which category is there **no** evidence of poor health?

- (A) 2 only (B) 3 only (C) 5 only (D) 6 only

24 Which position in the waterway (Q, R, S or T) is most likely to have the best health rating?

- (A) Q (B) R (C) S (D) T

For questions 25 and 26 use the information below.

Mud bricks are made from a mixture of clay, sand and straw. The mixture is poured into a wooden mould until it becomes firm. The moulded bricks are then turned out of the moulds and dried over time in the sun.

The table gives some information about mud bricks made by some students. Each student used the same amount of water and made the same amount of brick mixture.

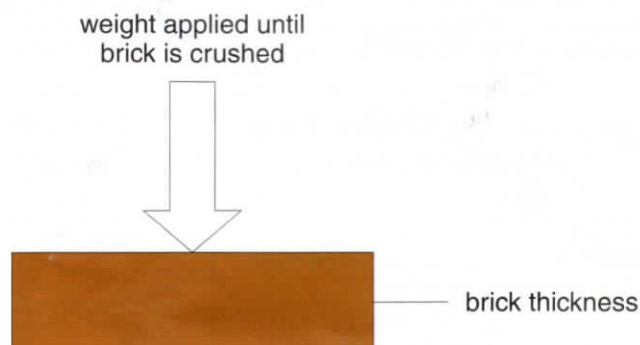
Student	Composition by volume (%)			Brick making Information			
	Clay	Sand	Straw	Straw length (cm)	Size (cm) (length $\times$ width $\times$ thickness)	Drying time (days)	Turning time (hours)
Jan	30	40	30	5	$50 \times 20 \times 15$	20	24
John	15	70	15	5	$50 \times 20 \times 15$	20	24
Mary	15	70	15	10	$50 \times 30 \times 15$	20	24
Matt	30	40	30	10	$50 \times 20 \times 15$	20	12

After 10 days Jan's bricks were dry on both sides, but Matt's bricks were dry on one side only.

25 What would most likely account for this difference?

- (A) Jan used more sand to make each brick than Matt.
- (B) Jan used less water to make her bricks than Matt.
- (C) Matt's bricks were thicker than Jan's bricks.
- (D) Matt turned his bricks so that one side never faced the sun.

26 The diagram shows how the students tested their mud bricks to see which one was strongest.



Mary's bricks were the strongest.

Which characteristic of the strongest brick distinguishes it from the other bricks?

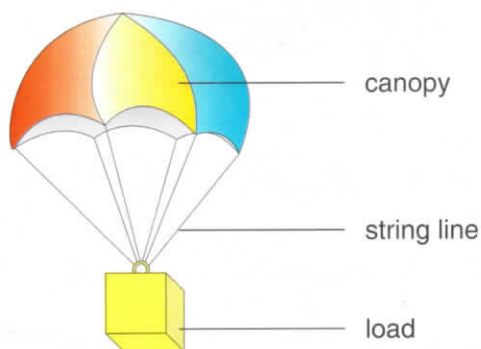
- (A) brick width
- (B) drying time
- (C) straw length
- (D) brick composition

27

Students investigated the effectiveness of various parachute designs. The table gives their experimental results.

Trial	Material used in canopy	Diameter of material used for canopy (cm)	Length of string lines (cm)	Time to fall 10 m (seconds)
1	tissue paper	20	30	25
2	tissue paper	30	30	30
3	tissue paper	30	40	28
4	cotton	20	30	20
5	cotton	20	30	18
6	cotton	30	30	28
7	plastic	20	40	23
8	plastic	30	40	28
9	plastic	30	30	30

The diagram shows the basic parts of their parachute designs.



Which trials provided a fair test of the effect of the material used in the canopy on the time to fall 10 metres?

- (A) 1, 4 and 7 only
- (B) 2, 6 and 9 only
- (C) 3, 6 and 8 only
- (D) 3, 7 and 8 only

28

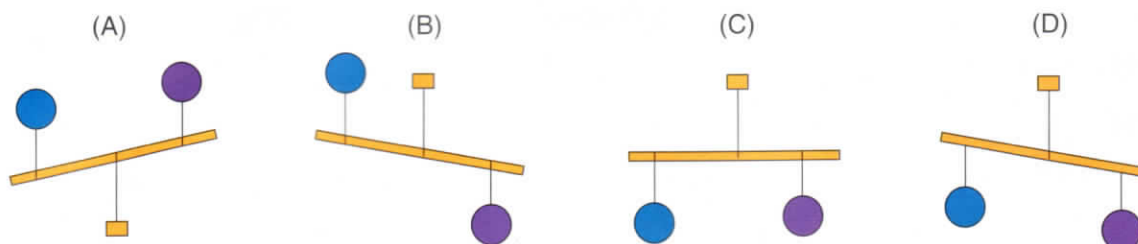
All gases have weight and take up space. If a gas is much lighter than air, a balloon filled with the gas will float up through air. If a gas is much heavier than air, a balloon filled with the gas will sink down through air.

The table compares the weight of some gases.

Some students used balloons containing the same amount of each of the gases and a long stick to compare their weights as shown.

hydrogen	lightest heaviest
helium	
nitrogen	
air	
oxygen	
carbon dioxide	

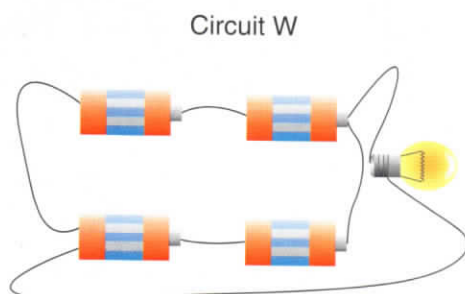
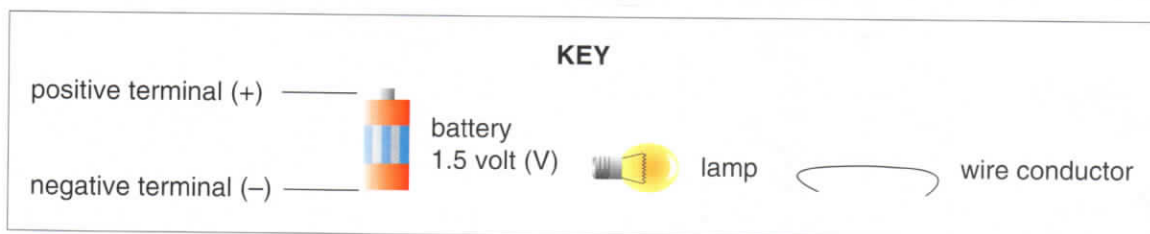
Which drawing correctly shows the result when the gases, helium and carbon dioxide, were compared?



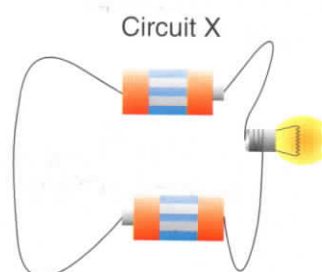
For questions 29 and 30 use the information below.

When batteries of the same voltage are joined in series, they produce a voltage equal to the sum of their individual voltages. When single batteries of the same voltage are joined in parallel, they produce the voltage of a single battery for a longer time.

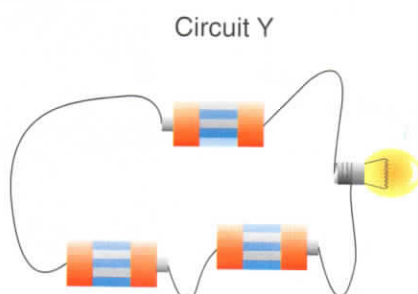
The diagrams show electrical circuits using 1.5 volt (V) batteries combined in series and in parallel that make lamps of different voltage glow for different lengths of time.



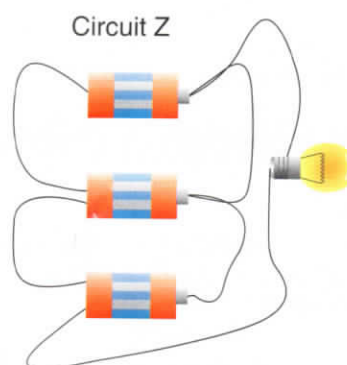
lamp glows for 2 hours



lamp glows for 1 hour



lamp glows for 1 hour



lamp glows for 3 hours

29 Which two circuits use batteries joined in parallel?

- (A) W and X (B) W and Z (C) X and Y (D) X and Z

30 Which circuit has the highest voltage across the lamp?

- (A) W (B) X (C) Y (D) Z

Year 5 - ICAS Science

2003 - Paper B

- | | |
|-------|-------|
| 1. C | 16. D |
| 2. C | 17. D |
| 3. D | 18. A |
| 4. A | 19. B |
| 5. B | 20. A |
| 6. D | 21. D |
| 7. C | 22. D |
| 8. B | 23. C |
| 9. A | 24. B |
| 10. D | 25. D |
| 11. D | 26. A |
| 12. B | 27. B |
| 13. C | 28. B |
| 14. C | 29. B |
| 15. B | 30. C |

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SOURCES

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<http://mearns.org.uk/mrsmith/bees/key.htm>
- Question 7** Bee graph from 53rd Conference Proceedings (2000) of the New Zealand Plant Protection Society Inc.
<http://www.hortnet.co.nz>
- Question 10** Image of crater courtesy of NASA and the Lunar and Planetary Institute.
<http://www.lpi.usra.edu/expmoon/science/craterstructure.html>
- Question 14** Image of ant copyright © Australian Museum.
http://wildlife.faunanet.gov.au/factfile.cfm?Fact_ID=248